Autoencoder

"Manifold hypothesis" – data lives on a low-dimensional manifold.

Dimensionality reduction

$x \in \mathbb{R}^D \mapsto z \in \mathbb{R}^d \qquad d < D \ (d \ll D)$

$f(x) = z$

- f – linear $\quad X \in \mathbb{R}^{N \times D}$ $\quad N$ – number of training objects

$Z \in \mathbb{R}^{N \times d} \qquad Z = X \cdot \Theta \qquad \Theta \in \mathbb{R}^{D \times d}$

Reconstruction

$\hat{X} = Z\,\Phi \qquad \Phi \in \mathbb{R}^{d \times D}$

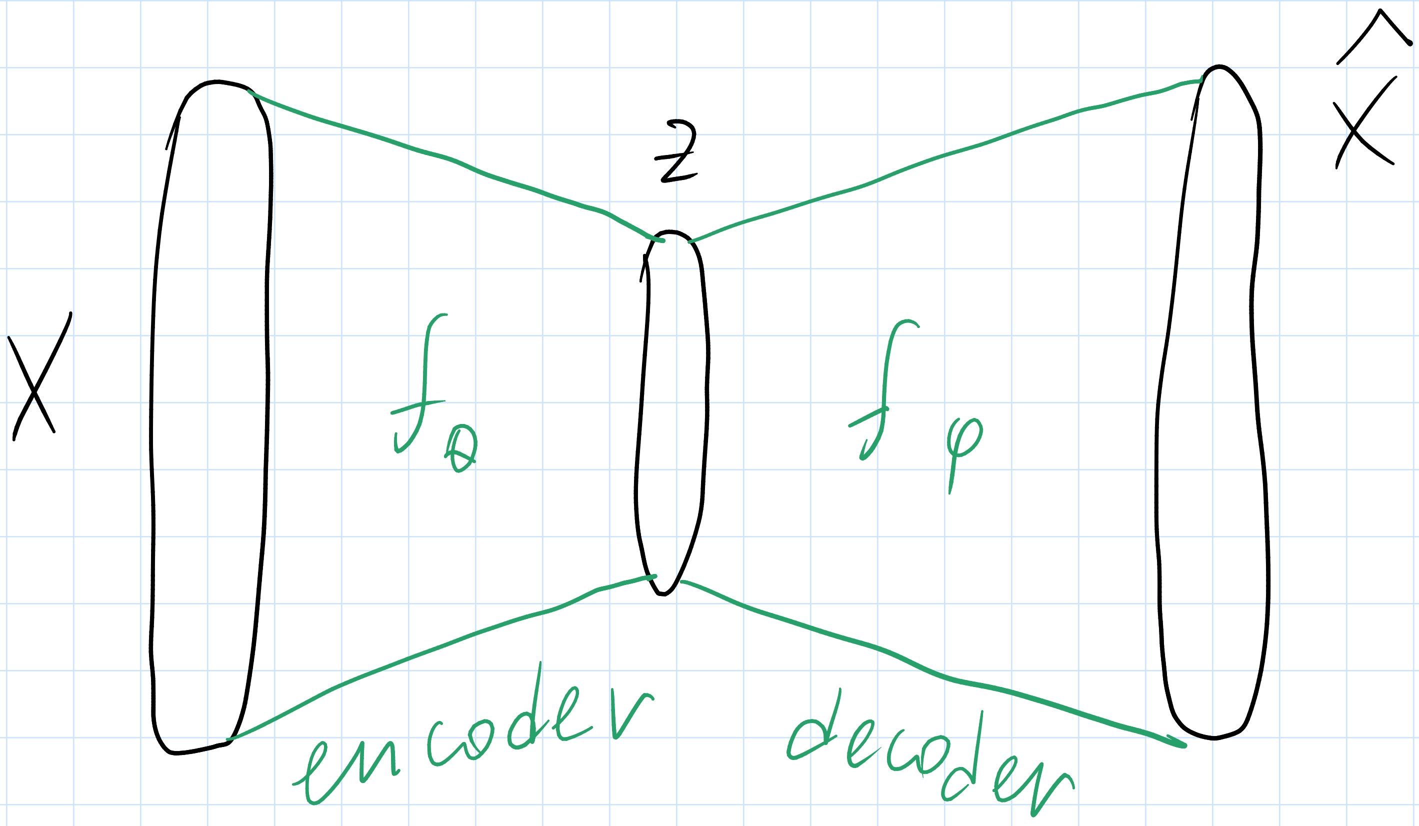
$$X \approx \hat{X} = X\,\Theta\,\Phi$$

$$MSE = \| X - \hat{X} \|_F^2 = \| X - X \underbrace{\Theta}_{D \times d} \underbrace{\Phi}_{d \times D} \|_F^2 \to \min_{\Theta, \Phi}$$

$$\|A\|_F = \sqrt{\sum_{ij} |A_{ij}|^2}$$

– Principal Component Analysis (PCA)

- f – non-linear

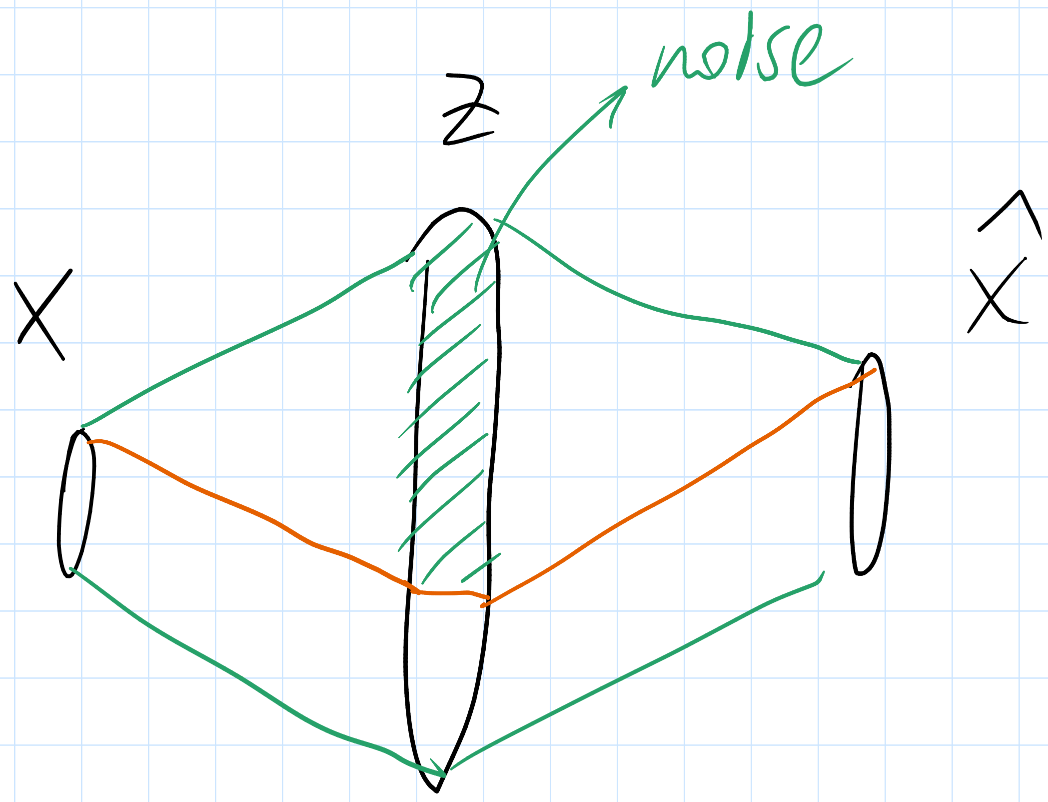


$\mathcal{X}$ – data distribution

$$\mathbb{E}_{X\sim\mathcal{X}} \|X-\hat{X}\|_2^2 = \mathbb{E}_{X\sim\mathcal{X}} \|X - f_\varphi(f_\theta(X))\|_2^2$$

$$\longrightarrow \min_{\varphi,\theta}$$

$d > D$:



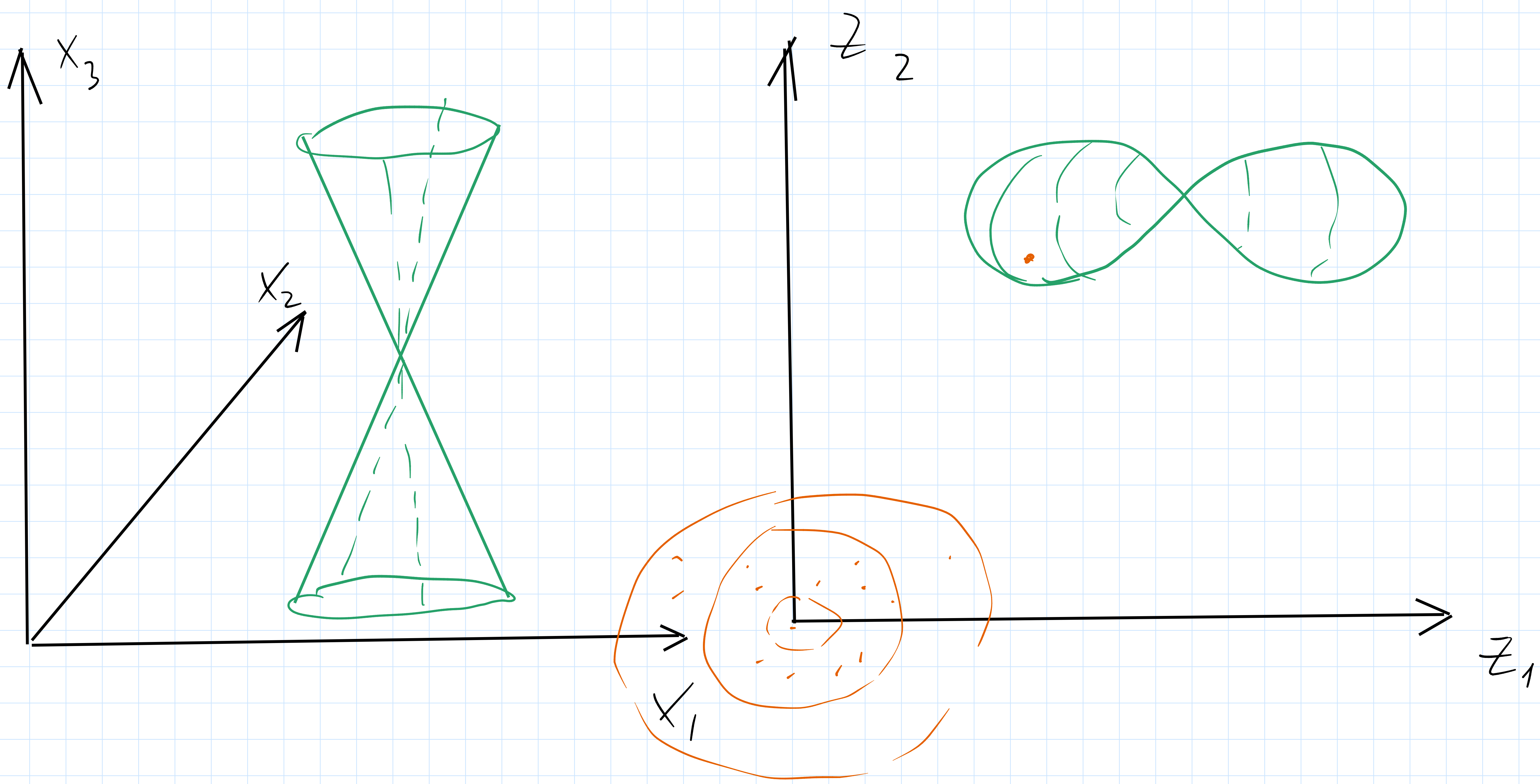
Denoising Autoencoder

$$\mathbb{E}_{X\sim\mathcal{X}} \|X - f_\varphi(f_\theta(\tilde{X}))\|_2^2 \rightarrow \min_{\varphi,\theta}$$

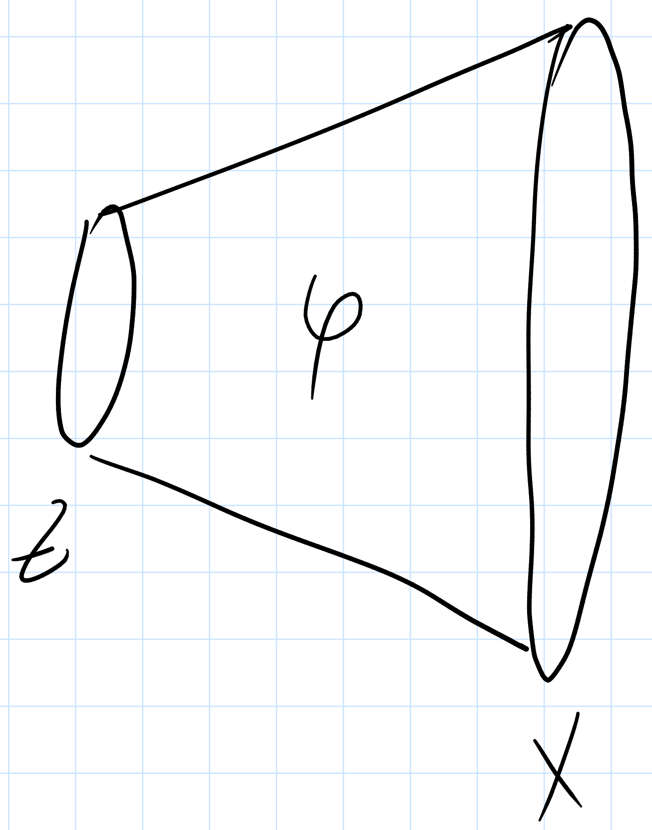
$$\tilde{X} = X + \varepsilon \qquad \varepsilon \sim N(0, \sigma^2 I)$$

Sampling from Autoencoder

$z \sim \underline{\mathcal{N}}(0, I) \qquad X_{gen} = f_\varphi(z)$



Variational Autoencoder (VAE)



$$p(x, z \mid \varphi) = p(x \mid z, \varphi)\, p(z)$$

$$p(x \mid z, \varphi) = \mathcal{N}(x \mid \underbrace{\mu_\varphi(z)}_{\in \mathbb{R}^D}, \underbrace{\Sigma_\varphi(z)}_{\in \mathbb{R}^{D \times D}})$$

$$p(z) = \mathcal{N}(z \mid 0, I)$$

x — observe $\quad z$ — latent

$$\log p(x \mid \varphi) = \log \int p(x, z \mid \varphi)\, dz \to \max_\varphi$$

Evidence lower bound (ELBO)

$$\log p(x|\varphi) = \log p(x|\varphi)\cdot 1 = \left\{ q(z) - \text{arbitrary distribution over } z \right\} =$$

$$= \log p(x|\varphi) \int q(z)\,dz = \int \log p(x|\varphi)\, q(z)\,dz =$$

$$= \left\{ p(x,z|\varphi) = p(z|x,\varphi)\, p(x|\varphi) \right\} =$$

$$= \int q(z) \log \frac{p(x,z|\varphi)}{p(z|x,\varphi)}\,dz = \int q(z) \log \frac{p(x,z|\varphi)\, q(z)}{p(z|x,\varphi)\, q(z)}\,dz =$$

$$= \underbrace{\int q(z) \log \frac{p(x,z|\varphi)}{q(z)}\,dz}_{} + \underbrace{\int q(z) \log \frac{q(z)}{p(z|x,\varphi)}\,dz}_{\parallel \atop KL(q\|p(z|x,\varphi))} =$$

$$= \underbrace{\mathcal{L}(q,\varphi)}_{\text{ELBO}} + KL(q\|p(z|x,\varphi))$$

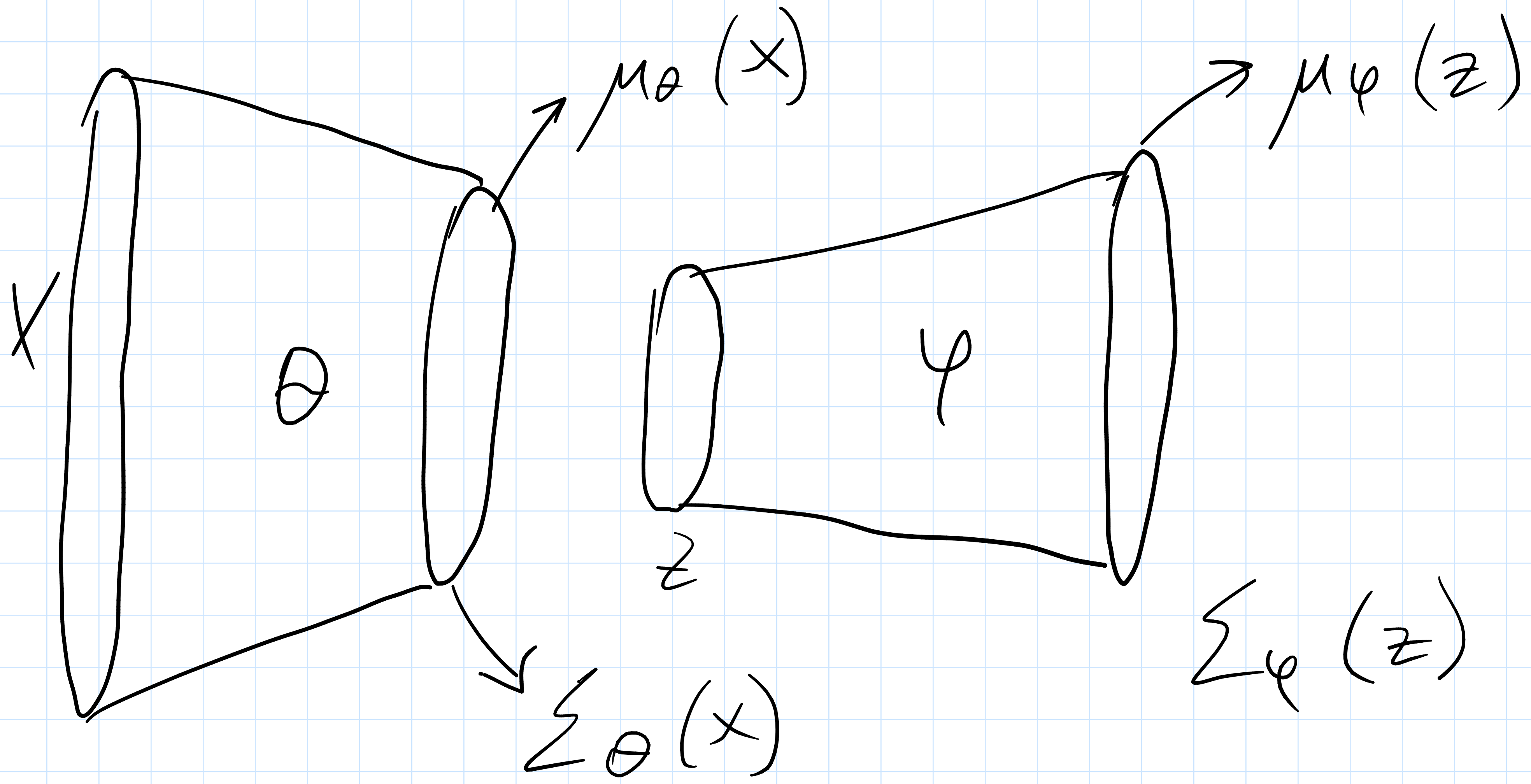
$q = p(z|x,\varphi)$ $\quad KL = 0 \quad$ $\mathcal{L}(q,\varphi) = \log p(x|\varphi)$

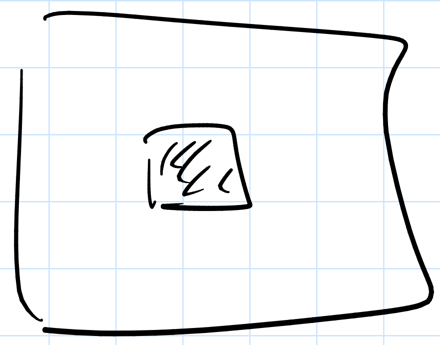
$q := q(z|x,\theta)$ — encoder

$p(x|z,\varphi)$ — decoder

$$\mathcal{L}(\theta,\varphi) = \int q(z|x,\theta) \log \frac{p(x,z|\varphi)}{q(z|x,\theta)}\,dz \to \max_{\varphi,\theta}$$

$$q(z|x,\theta) = \mathcal{N}(z|\mu_\theta(x), \Sigma_\theta(x))$$





$$\mu_\theta(x) \in \mathbb{R}^d \qquad \mu_\varphi(z) \in \mathbb{R}^D$$

$$\Sigma_\theta(x) = \mathrm{diag}(\sigma^2_{\theta_1}(x), \dots, \sigma^2_{\theta_d}(x))$$

$$\Sigma_\varphi(z) = \mathrm{diag}(\sigma^2_{\varphi_1}(z), \dots, \sigma^2_{\varphi_D}(z))$$

$$\Sigma_\varphi(z) = I$$

$$\mathcal{L}(\theta, \varphi) = \int q(z|x,\theta) \log \frac{p(x,z|\varphi)}{q(z|x,\theta)} dz =$$

$$= \{p(x,z|\varphi) = p(x|z,\varphi)\,p(z)\} = \int q(z|x,\theta) \log \frac{p(x|z,\varphi)\,p(z)}{q(z|x,\theta)} dz = \int q(z|x,\theta) \log p(x|z,\varphi) dz$$

$$+\int q(z|x,\theta)\log\frac{p(z)}{q(z|x,\theta)}\,dz =$$

$$=\underbrace{\mathbb{E}_{q(z|x,\theta)}\log p(x|z,\varphi)}_{\text{(MSE) reconstruction loss}} - \underbrace{KL\left(q(z|x,\theta)\,\|\,p(z)\right)}_{\text{regularization}} \to \max_{\varphi,\theta}$$

$$\mathbb{E}_{q(z|x,\theta)}\log p(x|z,\varphi) \to \nabla_\varphi,\ \nabla_\theta$$

$$\nabla_\theta\, \mathbb{E}_{q(z|x,\theta)}\log p(x|z,\varphi)$$

Reparametrization trick

$$\xi \sim N(\xi\,|\,m, s^2) \qquad \varepsilon \sim N(\varepsilon\,|\,0,1)$$

$$\xi = m + s\cdot\varepsilon$$

$$z \sim N\left(z\,|\,\mu_\theta(x),\ \Sigma_\theta(x)\right)$$

$$\varepsilon \sim N(\varepsilon\,|\,0, I) \qquad z = \mu_\theta(x) + \Sigma_\theta^{1/2}(x)\cdot\varepsilon$$

$$\nabla_\theta\, \mathbb{E}_{q(z|x,\theta)}\log p(x|z,\varphi) = \nabla_\theta\, \mathbb{E}_{\varepsilon\sim N(0,I)}\log p(x|\underbrace{z}_{\parallel},\varphi)$$

$$z\left(\mu_\theta(x), \Sigma_\theta(x), \varepsilon\right)$$

$$\log p(x|z,\varphi) = \log\left[\frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}}\exp\left((x-\mu)^T\Sigma^{-1}(x-\mu)\right)\right] =$$

$$= \underbrace{\cancel{-\frac{D}{2}\log 2\pi}}_{\text{const}} - \frac{1}{2}\log|\Sigma| + \frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu) =$$

$$= -\frac{1}{2}\sum_{i=1}^{D}\log\sigma_i^2 + \sum_{i=1}^{D}\frac{(x_i-\mu_i)^2}{\sigma_i^2} = \{\Sigma_\varphi(z) = I\} \propto$$

$$\propto \frac{1}{2}\sum_{i=1}^{D}(\underset{\substack{\uparrow \\ \text{object}}}{x_i} - \underset{\substack{\uparrow \\ \text{decoder output}}}{\mu_i})^2$$

$$q(z|x,\theta) = \mathcal{N}(z|\mu_\theta, \Sigma_\theta) \qquad p(z) = \mathcal{N}(z|0,I)$$

$$\xi_1 \sim \mathcal{N}(\xi_1|\mu_1,\Sigma_1) \qquad \xi_2 \sim \mathcal{N}(\xi_2|\mu_2,\Sigma_2)$$

$$KL(\xi_1\|\xi_2) = \frac{1}{2}\left[tr(\Sigma_2^{-1}\Sigma_1) - d + (\mu_2-\mu_1)\Sigma_2^{-1}(\mu_2-\mu_1)\right.$$

$$\left. + \log\frac{|\Sigma_2|}{|\Sigma_1|}\right]$$

$$\Sigma_\theta - diag(\sigma_1^2 \ldots \sigma_d^2) \qquad \Sigma_p = I$$

$$KL(q\|p) = \frac{1}{2}\left[tr\,\Sigma_\theta - d + \|\mu_\theta\|_2^2 - \log|\Sigma_\theta|\right] =$$

$$= \frac{1}{2}\sum_{i=1}^{d}(\sigma_i^2 + \mu_i^2 - \log\sigma_i^2) - \frac{d}{2}$$

$\log \sigma^2$ — output of layer

$$\sigma_i^2 = \exp(\log \sigma_i^2)$$